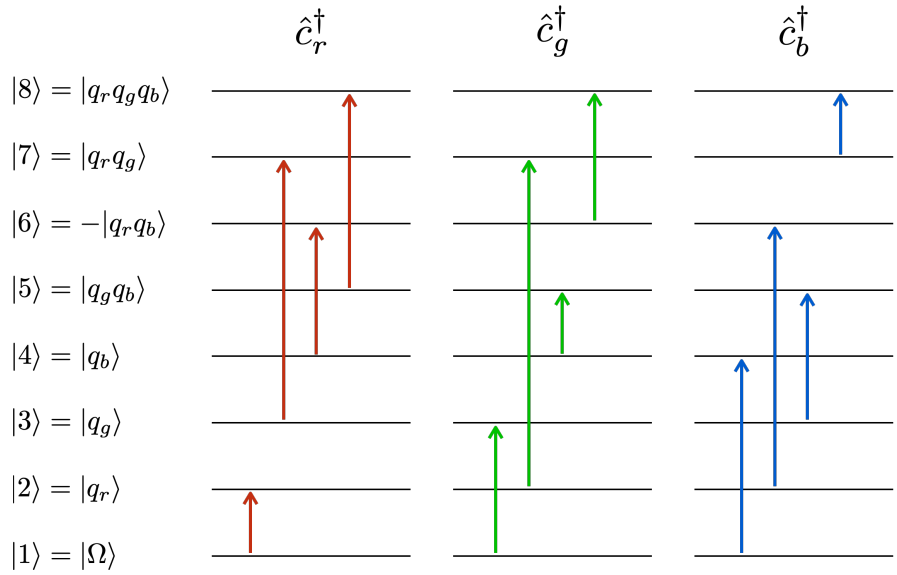

FIG. 2. Transitions among the quark states mapped to a qu8it, defined in Eq. (6), induced by the creation operators $\hat{c}_r^\dagger$ (left), $\hat{c}_g^\dagger$ (center) and $\hat{c}_b^\dagger$ (right) from Eq. (13). The color of the arrows corresponds to the color charge of the operator.

IV. THE 1+1D QCD HAMILTONIAN MAPPED TO QU8ITS

In terms of the annihilation and creation operators, the $N_f = 1$ SU(3) Hamiltonian in Eq. (1) can be written as

$$\begin{aligned} \hat{H} &= \hat{H}_{kin} + \hat{H}_m + \hat{H}_{el} \\ &= \frac{1}{2} \sum_{n=0}^{2L-2} \sum_{\alpha=r,g,b} \left(\hat{c}^\dagger_{\alpha,n} \hat{c}^\dagger_{\alpha,n+1} - \hat{c}_{\alpha,n} \hat{c}_{\alpha,n+1} \right) + 3m \sum_{n=0}^{2L-1} \hat{\mathcal{B}}_n + \frac{g^2}{2} \sum_{n=0}^{2L-2} \sum_{a=1}^{8} \left(\sum_{m \leq n} \hat{Q}^{(a)}_m \right)^2 . \end{aligned} \tag{15}$$

When written explicitly in terms of matrix-operators acting on qu8its (analogous to the JW mapping to qubits with the operators written in terms of Pauli matrices), it becomes

$$\begin{aligned} \hat{H} \to \frac{1}{2} \sum_{n=0}^{2L-2} \sum_{\alpha=r,g,b} \left[\left(\widetilde{c_\alpha}^\dagger \, \widetilde{P} \right)_n \otimes \widetilde{c_{\alpha,n+1}}^\dagger - \left(\widetilde{c_\alpha} \, \widetilde{P} \right)_n \otimes \widetilde{c_{\alpha,n+1}} \right] + 3m \sum_{n=0}^{2L-1} \widetilde{\mathcal{B}}_n \\ + \frac{g^2}{2} \sum_{n=0}^{2L-2} \sum_{a=1}^{8} \left(\sum_{m \in \text{even}}^{n} \widetilde{Q}^{(a)}_m + \sum_{m \in \text{odd}}^{n} \widetilde{\bar{Q}}^{(a)}_m \right)^2 , \end{aligned} \tag{16}$$

where the phase matrix $\widetilde{P}$ is

$$\hat{P} \to \widetilde{P} = \text{diag}\,(1, -1, -1, -1, 1, 1, 1, -1) \ . \tag{17}$$

The $\widetilde{P}$ matrix is the generalized form of the σ^z Pauli matrix, generally acting on strings of qudits between quark operators in order to satisfy Fermi statistics. Examining the connectivity map shown in Fig. 3, one observes that each state is connected to either 3 or 5 other states within the qu8it. The states corresponding to color-singlet states have connectivity of 3 (originating from the kinetic term), while those corresponding to color-triplet or anti-triplet states have connectivity of 5 (originating from the kinetic and chromo-electric terms).

The extension to systems with a larger number of quark flavors, $N_f > 1$, is straightforward. For each flavor of quark at a given lattice site, there is a corresponding qu8it, and similarly for anti-quark sites. The kinetic and mass terms in the Hamiltonian are replicated for each flavor (with the appropriate masses), and the color-charge operators are extended as in Eq. (2).

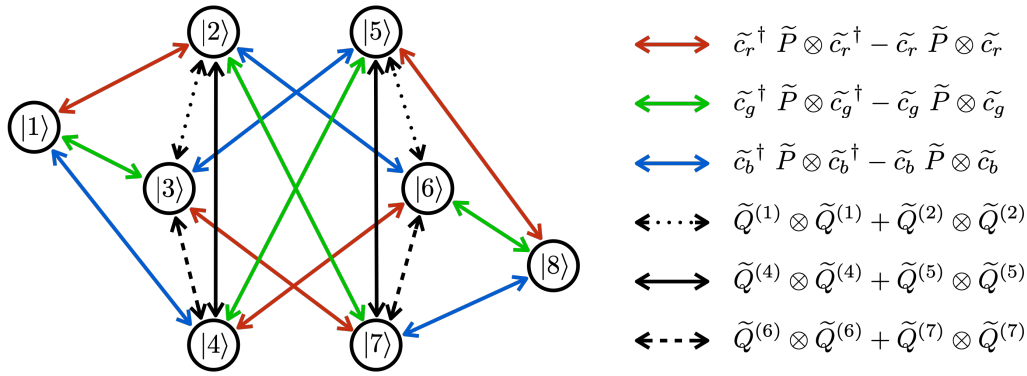

FIG. 3. A connectivity map among the eight qu8it states that is required by the Hamiltonian in Eq. (16). Colored connections are for the kinetic term, while black ones are for the color charge-charge interactions (different line styles correspond to different charge combinations).

A. QCD with $N_f = 1$ on $L = 1$ Spatial Site with OBCs

It is helpful to explore examples of mappings to qu8its. Consider $L = 1$, with lattice sites $n = 0, 1$, with one flavor $N_f = 1$, which maps to two qu8its (one for the quarks and one for the anti-quarks), a system that we have studied previously [37]. The matrix representation of the Hamiltonian, as given in Eq. (16), reduces to

$$\begin{aligned} H_1 \;=\; & \frac{1}{2}\left(\widetilde{c}_r^\dagger\ \widetilde{P}\otimes\widetilde{c}_r^\dagger + \widetilde{c}_g^\dagger\ \widetilde{P}\otimes\widetilde{c}_g^\dagger + \widetilde{c}_b^\dagger\ \widetilde{P}\otimes\widetilde{c}_b^\dagger - \widetilde{c}_r\ \widetilde{P}\otimes\widetilde{c}_r - \widetilde{c}_g\ \widetilde{P}\otimes\widetilde{c}_g - \widetilde{c}_b\ \widetilde{P}\otimes\widetilde{c}_b\right) \\ & + 3m\ \left(\widetilde{\mathcal{B}}\otimes\widetilde{I} + \widetilde{I}\otimes\widetilde{\mathcal{B}}\right) \;+\; \frac{g^2}{2}\sum_a \left(\widetilde{Q}^{(a)}\otimes\widetilde{I}\right)^2 \;+\; \frac{h^2}{2}\sum_a \left(\widetilde{Q}^{(a)}\otimes\widetilde{I} + \widetilde{I}\otimes\widetilde{Q}^{(a)}\right)^2 , \\ =\; & H_{1kin} \;+\; H_{1m} \;+\; H_{1el} \;+\; H_{1h} \;, \end{aligned} \tag{18}$$

where $\widetilde{I}$ is the identity operator. The term with coefficient h has been included to enforce color-neutrality across the lattice as $h \to \infty$, as we implemented in previous work [37]. This generates a significant penalty for chromo-electric-energy density beyond the end of the spatial lattice, and without this term color-edge states appear as low-lying states in the spectrum due to OBCs [37]. In the large-h limit, only color-singlet states remain at low energies.

This system is sufficiently simple and of small dimensionality, involving a 64×64 Hamiltonian matrix, that it can be exactly diagonalized with classical computers. Projecting to states with good baryon number further reduces the size of the matrix. For example, in the $B = 0$ sector, the contributing configurations correspond to i) both qu8its in the vacuum (a $\mathbf{1}$); ii) the qu8its are in the one-quark one-anti-quark sector ($\mathbf{3}\otimes\overline{\mathbf{3}} = \mathbf{8}\oplus\mathbf{1}$); iii) the qu8its are in the two-quark two-anti-quark sector, ($\overline{\mathbf{3}}\otimes\mathbf{3} = \mathbf{8}\oplus\mathbf{1}$); and, iv) both qu8its are in the completely occupied state, a baryon-anti-baryon pair (a $\mathbf{1}$). Consequently, the total number of $B = 0$ basis states is $n_{B=0} = 1 + 9 + 9 + 1 = 20$. However, a large value of h propels the $\mathbf{8}$'s high in the spectrum, leaving only four color-singlet states in the low-lying spectrum. These are formed from linear combinations of the eight pairings of states in the qu8its.[7]

As this is a system we have analyzed previously using the JW mapping to qubits [37], the low-lying spectra and time-evolution from arbitrary initial states are known. The (exact) time evolution found from matrix exponentiation of the Hamiltonian in Eq. (18), is found to furnish results that agree with our previous analyses.

As shown in Eq. (21) below, the chromo-electric term $\hat{H}_{1el}$ is diagonal in the qu8it computational basis for $L = 1$. Thus, in the case of an ideal quantum computer, with an initial state that is a color-singlet, exact time evolution will leave the system in a color-singlet state at all subsequent times, even without the "h-term" in Eq. (18). As such, that term can be omitted in the time-evolution operator in the case of $L = 1$. For systems with $L > 1$, however, color-charge is violated by Trotterized time evolution (in particular, due to Trotterization of the eight contributions to the color sum in the chromo-electric field term, when the color-charge operators act on different sites), and, consequently, including the "h-term" is a means to mitigate this violation.

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[7] If we were working in U(1) lattice gauge theory describing quantum electrodynamics, the situation would be somewhat less complex because each (tensor-product) basis state is an eigenstate of the electric-charge operator. This is not the situation for non-Abelian theories, where the color-charge operator generally mixes basis states.

1. Quantum Circuits and Givens Rotations

To establish the quantum circuits for the $N_f = 1$ and $L = 1$ system, the Hamiltonian is decomposed into unitary operations on each of the qu8its. As is standard for quantum operations on qudits, the Hamiltonian in Eq. (18) is decomposed into generators of Givens rotations, $\mathcal{X}_{ij}$, $\mathcal{Y}_{ij}$, and $\mathcal{Z}_i$. In this case, $d = 8$, these provide a complete set of generators for SU(8) transformations on a single qu8it. It is convenient, in an effort to better connect with hardware aspects of simulations, to work with Hadamard-Walsh matrices, w_i, rather than with $\mathcal{Z}_i$. For $d = 8$, there are 28 $\mathcal{X}_{ij}$, 28 $\mathcal{Y}_{ij}$, and 8 w_i (including the identity), as defined in App. C.

The kinetic-energy operator in the Hamiltonian in Eq. (18) is comprised only of terms that act on both qu8its. As the annihilation and creation operators induce multiple transitions within a qu8it, as depicted in Fig. 2, this term decomposes into multiple tensor products of Givens matrices,

$$\begin{aligned} H_{1kin} &= \frac{1}{4} \sum_{\substack{r\in\{(12),(13),(14),\\ (58),(68),(78),\\ -(26),-(27),-(35),\\ -(37),-(45),-(46)\}}} (\mathcal{X}_r \otimes \mathcal{X}_r - \mathcal{Y}_r \otimes \mathcal{Y}_r) \\ &+ \frac{1}{4} \sum_{\substack{(r,s)\in\{(12)(58),(13)(68),\\ (14)(78),(26)(35),\\ (27)(45),(37)(46)\}}} (\mathcal{X}_r \otimes \mathcal{X}_s + \mathcal{X}_s \otimes \mathcal{X}_r - \mathcal{Y}_r \otimes \mathcal{Y}_s - \mathcal{Y}_s \otimes \mathcal{Y}_r) \\ &+ \frac{1}{4} \sum_{\substack{(r,s)\in\{(12)(37),(46)(12),\\ (27)(13),(13)(45),(14)(26),\\ (35)(14),(78)(26),(27)(68),\\ (35)(78),(58)(37),(68)(45),\\ (46)(58)\}}} (\mathcal{X}_r \otimes \mathcal{X}_s - \mathcal{X}_s \otimes \mathcal{X}_r - \mathcal{Y}_r \otimes \mathcal{Y}_s + \mathcal{Y}_s \otimes \mathcal{Y}_r) \ , \end{aligned} \tag{19}$$

where the minus sign in some of the indices in the first summation means a global minus sign for that corresponding term. The mass term in the Hamiltonian is diagonal, given in terms of the action of the baryon-number matrix $\widetilde{\mathcal{B}}$ defined in Eq. (12), and which can be further decomposed into the w_i,

$$\begin{aligned} H_{1m} &= 3m \left(\widetilde{\mathcal{B}} \otimes \widetilde{I} + \widetilde{I} \otimes \widetilde{\mathcal{B}}\right) \\ &= \frac{m}{\sqrt{2}} \left((6w_1 - 3w_2 - w_4 - w_6 - w_8) \otimes \widetilde{I} \ + \ \widetilde{I} \otimes (6w_1 - 3w_2 - w_4 - w_6 - w_8)\right) \ , \\ \widetilde{\mathcal{B}} &= \frac{1}{3}\mathrm{diag}\,(0,1,1,1,2,2,2,3) \ = \ \frac{1}{3\sqrt{2}}(6w_1 - 3w_2 - w_4 - w_6 - w_8) \ . \end{aligned} \tag{20}$$

For a single site, the contribution of the chromo-electric-energy density to the Hamiltonian from within the lattice, receiving contributions from the $n = 0$ qu8it site only, can be written as (details can be found in App. D),

$$\begin{aligned} H_{1el} &= \frac{g^2}{2} \sum_a \left(\widetilde{Q}^{(a)} \otimes \widetilde{I}\right)^2 \ \rightarrow \ \frac{2g^2}{3}\mathrm{diag}\,(0,1,1,1,1,1,1,0) \otimes \widetilde{I} \\ &= \frac{g^2}{2} \left(8w_1 \otimes w_1 - \frac{8}{3}\,(w_3 + w_5 + w_7) \otimes w_1\right) \\ &= \frac{g^2}{2} \left(\widetilde{I} \otimes \widetilde{I} - \mathrm{diag}\left(1, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, 1\right) \otimes \widetilde{I}\right) \ . \end{aligned} \tag{21}$$

The h-term introduced to enforce color neutrality of an $L > 1$ lattice, requires computing the square of the total color